Shark Tank India Dataset Analysis Report: by Ashutosh mishra

Introduction

This project analyzes the **Shark Tank India** dataset to uncover insights about startup pitches, deals, investors, and funding trends. The goal was to understand which startups received investments, how much funding was offered, and which sharks were the most active investors. The analysis focuses on deal patterns, sectoral trends, and the geographical spread of entrepreneurs featured on the show.

Abstract

The Shark Tank India dataset contains information about multiple seasons, episodes, and pitches. Each record includes details such as startup name, industry, city, deal amount, equity offered, and shark participation.
Through data cleaning, feature engineering, and visualization, the analysis provided insights into:

- The number of startups that received offers and accepted deals.
- The total and average investment amounts.
- The most active and generous investors (“sharks”).
- Industry sectors that attracted the most funding.
- City and state trends in entrepreneurship.

Tools used:

Programming Language: Python

Libraries: Pandas, NumPy, Matplotlib, Seaborn

Environment: Jupyter notebook

Visualization: Bar Charts, Heatmaps, Scatterplots

These tools were used for data manipulation, missing value handling, visualization, and deriving key insights.

Steps Involved in Building the Project

1. **Data Import & Inspection**
   The dataset was loaded using Pandas and examined for its shape, structure, and data types using df.info() and df.describe(include='all').
2. **Data Cleaning:**

- Missing investment-related fields were filled with 0.
- Categorical columns (e.g., *Industry*, *City*, *Sector*) were filled with “Unknown”.
- Date columns were converted into datetime format.
- Inconsistent entries (e.g., multiple cities/states per startup) were standardized by extracting the first value.

3. **Exploratory Data Analysis (EDA)**
    - Total and average deal amounts were calculated.
    - The most active sharks and most-funded industries were identified.
    - Graphs showed industry-wise investment patterns and city-wise participation.
    - Season and episode-level summaries were used to explore show growth trends.

4. **Visualization & Insights**
    Visualizations using Matplotlib and Seaborn provided:
    - Shark-wise total investments (bar plots).
    - Industry and city distributions of pitches.
    - Correlation between amount invested and equity percentage.

Conclusion

This analysis of Shark Tank India data provided valuable insights into the startup ecosystem showcased on the show.
Key findings include:

- Only a portion of pitches converted into actual deals.
- Certain industries like *Food & Beverage* and *Tech* dominated investments.
- A few sharks (notably Peyush, Aman, and Namita) invested in a higher number of startups.
- Startups from metro cities like Mumbai, Delhi, and Bengaluru received greater representation.

The project demonstrates practical skills in **data cleaning, feature engineering, visualization, and business insight generation**, making it a strong example of real-world data analysis work.
